

SE lab-3

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Architecture Selection

We chose **Microservices Architecture** for the Student Project Portfolio & Showcase App.

Reason 1 – Independent Scaling

The public voting and leaderboard functionality may experience high traffic during the project expo.

Microservices allow the Voting and Leaderboard services to be scaled independently without scaling the entire application.

Reason 2 – Fault Isolation

The application contains several independent functionalities such as project management, artifact management, evaluation, voting, and leaderboard calculation. Separating these into services helps isolate failures so that a problem in one service does not necessarily affect the entire application.

Security Advantage

A dedicated Authentication & Verification Service can handle user authentication and voter verification. The Voting Service can use this service to prevent duplicate or fraudulent voting, supporting the application's anti-sybil requirement.

Performance Benefit

Voting and leaderboard services can be independently optimized and scaled to handle high traffic and support the requirement for real-time updates with API response latency below 300ms.